Values variables and types

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Python readiness portfolio

THE CONNECTED WEEK

One connected types-and-variables record.

Explain values, trace a saved snapshot, diagnose operation errors and state a data-meaning limit.

Keep one habit: Keep type, current value, statement order, observed output and task role separate.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep the stated task, its own code/values and support record together.

Task and evidence record

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from From a paper algorithm to Python. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

MAIN LESSON · SECTION 5

W1 · Explain the procedure

Keep the stated task, its own code/values and support record together.

Task and evidence record

- Saved-value practice: count 8, saved 8; after count=count+2, count 10 and saved 8; a new saved=count makes saved 10. Picture count 8/tag "8": outputs 10 and text 82; count 10 changes only numeric output to 12. Mixed integer/string addition raises TypeError. int/float/str/bool describe values; task rules separately reject negative object counts or preserve a leading-zero identifier.

- Classify 12, 1.5, "12", False and "False". Explain why the last two are different.

My reasoning and evidence

MAIN LESSON · SECTION 6

W2 · Predict, trace and observe

> Keep the stated task, its own code/values and support record together.

Task and evidence record

- Saved-value practice: count 8, saved 8; after count=count+2, count 10 and saved 8; a new saved=count makes saved 10. Picture count 8/tag "8": outputs 10 and text 82; count 10 changes only numeric output to 12. Mixed integer/string addition raises TypeError. int/float/str/bool describe values; task rules separately reject negative object counts or preserve a leading-zero identifier.

- Reference for this exercise: open this week's student lesson, "Values variables and types", at "See how it works" and "Follow a worked example". Use the original worked values or code named in the question; the later pictorial case is a separate example. Keep this reference beside your workbook while practising.

- Trace the five-line count and saved program. Record both printed **output**s, then predict the result of assigning saved = count after the change. Check with a run or label a paper trace.

> **My reasoning and evidence**

MAIN LESSON · SECTION 7

W3 · Find the first error

> Keep the stated task, its own code/values and support record together.

Task and evidence record

- Saved-value practice: count 8, saved 8; after count=count+2, count 10 and saved 8; a new saved=count makes saved 10. Picture count 8/tag "8": outputs 10 and text 82; count 10 changes only numeric output to 12. Mixed integer/string addition raises TypeError. int/float/str/bool describe values; task rules separately reject negative object counts or preserve a leading-zero identifier.

- Predict 6 + 3, "6" + "3" and "6" * 3. Explain what happens for "6" + 3. Which expression counts six groups of three?

My reasoning and evidence

MAIN LESSON · SECTION 9

W5 · State an honest limit

> Keep the stated task, its own code/values and support record together.

Task and evidence record

- Saved-value practice: count 8, saved 8; after count=count+2, count 10 and saved 8; a new saved=count makes saved 10. Picture count 8/tag "8": outputs 10 and text 82; count 10 changes only numeric output to 12. Mixed integer/string addition raises TypeError. int/float/str/bool describe values; task rules separately reject negative object counts or preserve a leading-zero identifier.

- Give a suitable **type** for a counter count, a length of 2.5 metres, a shelf ID with a leading zero, and a yes-or-no availability flag. Give a count **value** with the right **type** that should still be rejected.

> **My reasoning and evidence**

MAIN LESSON · SECTION 4

Try alone · Selected independent task

> Keep the stated task, its own code/values and support record together.

Task and evidence record

- Predict both **output** lines and explain why estimate stays unchanged when count is reassigned. Repeat the trace with initial count=0. Explain why note is text even though its printed characters look like a number.

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Trace values, types and reassignment

```
count = 4
note = "4"
estimate = count / 2
fits = count <= 5
count = count + 3
print(count, note, estimate, fits)
print(type(count).__name__, type(note).__name__, type(estimate).__name__,
type(fits).__name__)
```

> **My independent explanation, trace and limits**
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MAIN LESSON · SECTION 10

Project · Keep code and evidence

> Keep the stated task, its own code/values and support record together.

Task and evidence record

- Add a labelled **value**s **variable**s and **type**s **evidence** sheet to Python readiness portfolio. Show how you can distinguish integer, floating-point, text and boolean **value**s.

  Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

MAIN LESSON · SECTION 11

Feedback · Repair and return

> Keep the stated task, its own code/values and support record together.

Task and evidence record

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep the stated task, its own code/values and support record together.

Task and evidence record

- Use W2: Trace a saved **value** for **Value**s **variable**s and **type**s. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep the stated task, its own code/values and support record together.

Task and evidence record

- Use the worked case for **Value**s **variable**s and **type**s. Proposed explanation: “Digits always mean a number”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Additional practice · Earlier W4 transfer

REFERENCE · SECTION 8

Additional practice · Earlier W4 transfer

> Keep the stated task, its own code/values and support record together.

Task and evidence record

- A fresh script runs seats = 12; booked = seats; seats = seats - 5; print(seats); print(booked), with each statement on a separate line. Predict both **output**s. Then add booked = seats + 1 and print(booked).

  Explain the evaluation order and choose a **type** for a seat label "07".

> **My reasoning and evidence**

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 12

Extra practice · Pictured case

> Keep the stated task, its own code/values and support record together.

Task and evidence record

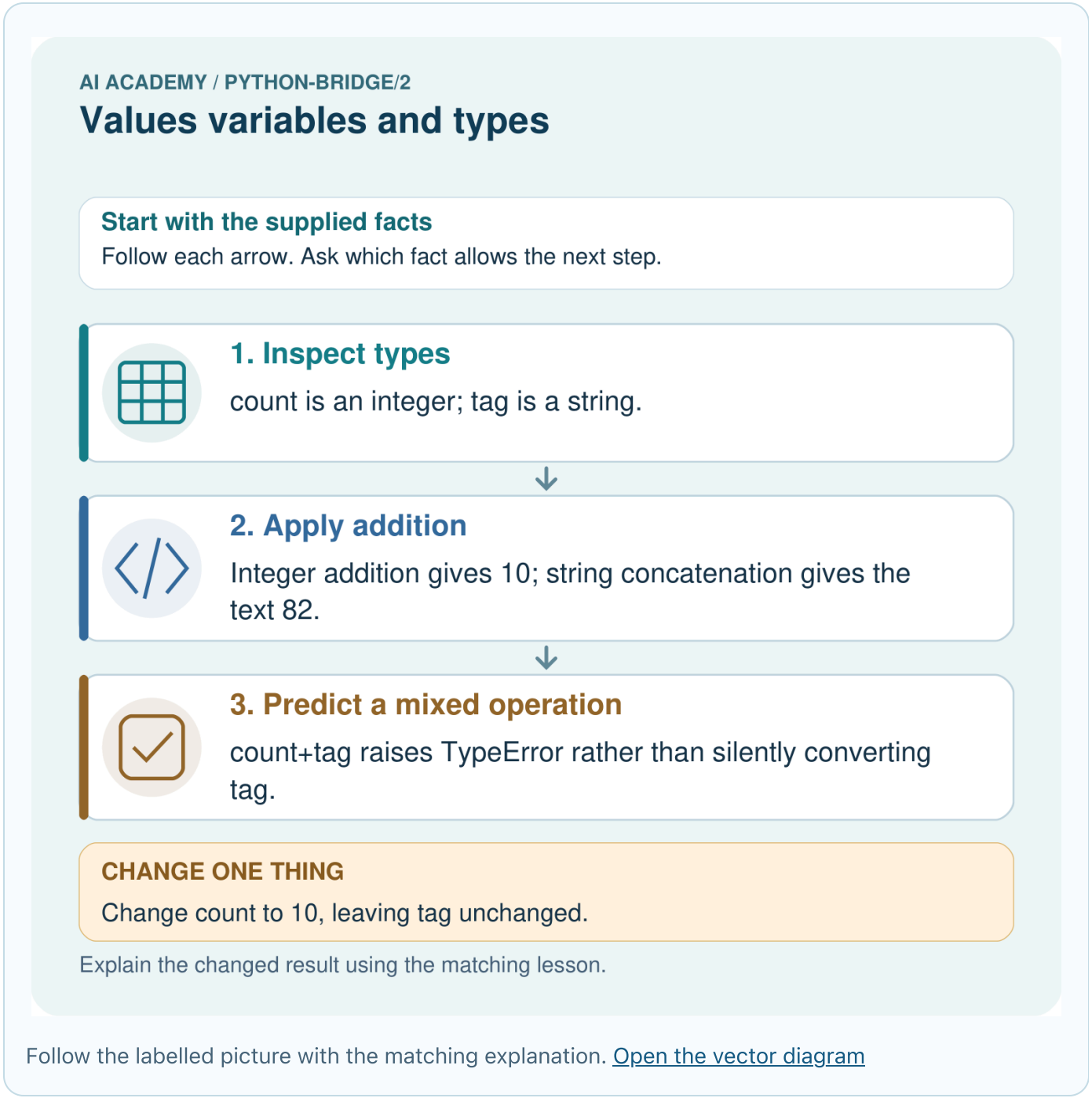
- A number and its printed-looking text. A program stores a numeric badge count and a **string** that looks like that count.

- Python 3: count=8; tag="8"; print(count+2); print(tag+"2").

- Predict both print **output**s and explain why they differ. Does count+tag work, and what happens to each valid **output** after count becomes ten?

- Support used: alone / hint / worked example. This record is practice **evidence**.
- Project connection: For Python readiness before the Responsible Plant Classifier, retain a trace of the **value**s, the predicted **output**s and the changed-case explanation.

  Use actual Python 3 results to check the **prediction** after reasoning first.



Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

▼ **Reference · Project portfolio milestone**

REFERENCE · SECTION 13

Reference · Project portfolio milestone

> Keep the stated task, its own code/values and support record together.

Task and evidence record

- Project: Python readiness portfolio
- Current milestone: Python readiness portfolio. Keep runnable code, **prediction**s, actual **output**s, one repaired error and an explanation of a function on changed **input**s.
- For Python readiness before the Responsible Plant Classifier, retain a trace of the **value**s, the predicted **output**s and the changed-case explanation. Use actual Python 3 results to check the **prediction** after reasoning first.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	

Evidence to keep	My record
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ **Extra practice · Predict, check and repair**

REFERENCE · SECTION 14

Extra practice · Predict, check and repair

> Use only after an attempt; keep the reserved assessment separate.

First question and hint

- A number and its printed-looking text. A program stores a numeric badge count and a **string** that looks like that count.
- Which labelled step matches this detail? count+tag raises **Type**Error rather than silently converting tag.
- 1. Predict a mixed operation
- 2. Inspect **type**s

- 3. Apply addition

- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**
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▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Predict a mixed operation count+tag raises **Type**Error rather than silently converting tag. **Value**s carry **type**s, not merely appearances.

The plus operator performs numeric addition for these integers and concatenation for these **string**s; the mixed operation is invalid here.

Changed-case question and hint

- Changed-case practice: Change count to 10, leaving tag unchanged. Which conclusion is supported by the supplied facts?

- 1. This one example guarantees correct results for every future **input**.

- 2. We can decide the result without examining the changed **input** or the procedure.

- 3. The two valid **output**s become 12 and 82.

- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the changed explanation**

After your attempt, compare with the supported explanation: The two valid **output**s become 12 and 82. Only the integer **variable** changes; the **string** still contains the character 8.

Repair and keep the record

Review From a paper algorithm to Python (python-bridge/1). Goal: Translate a short paper calculation into a Python script. Short practice: Match each paper quantity to its Python name.

Why does the spare count get added once, and what prints when spare is five?. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

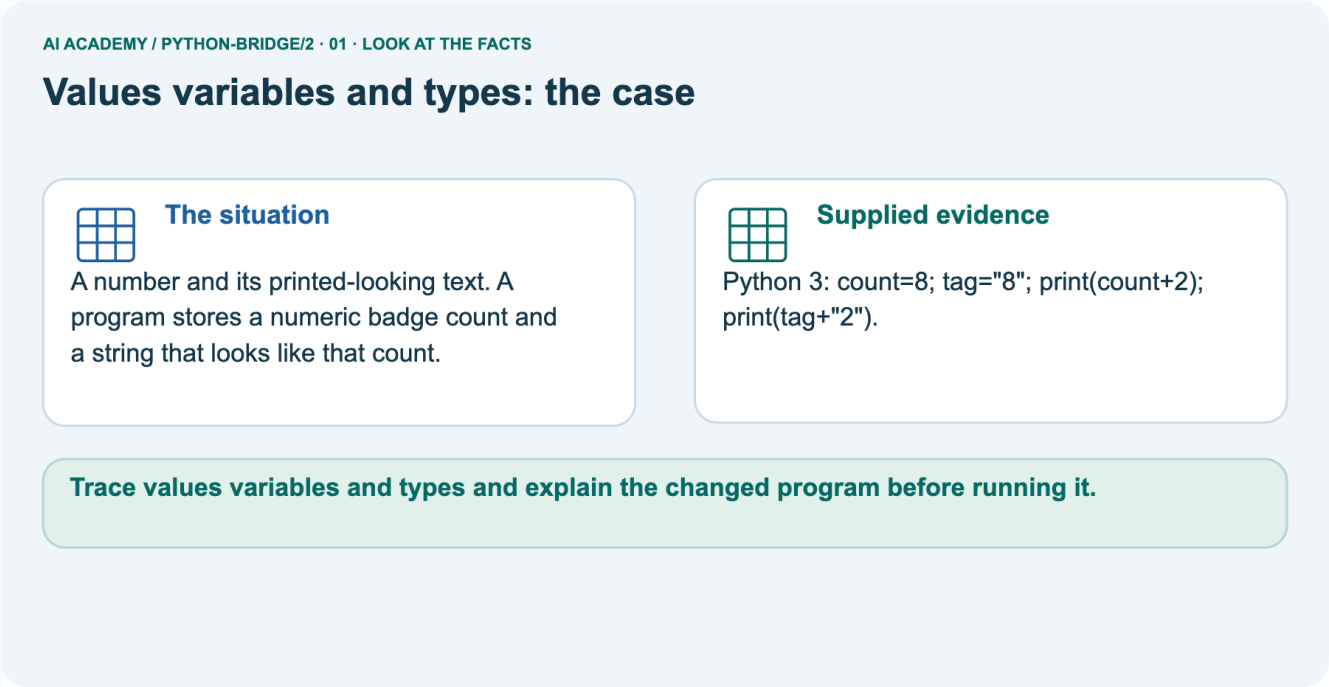
▼ Reference · Six-picture response sheets

REFERENCE · SECTION 15

Reference · Six-picture response sheets

> Use the picture relevant to a current gap.

01. Values variables and types: the case



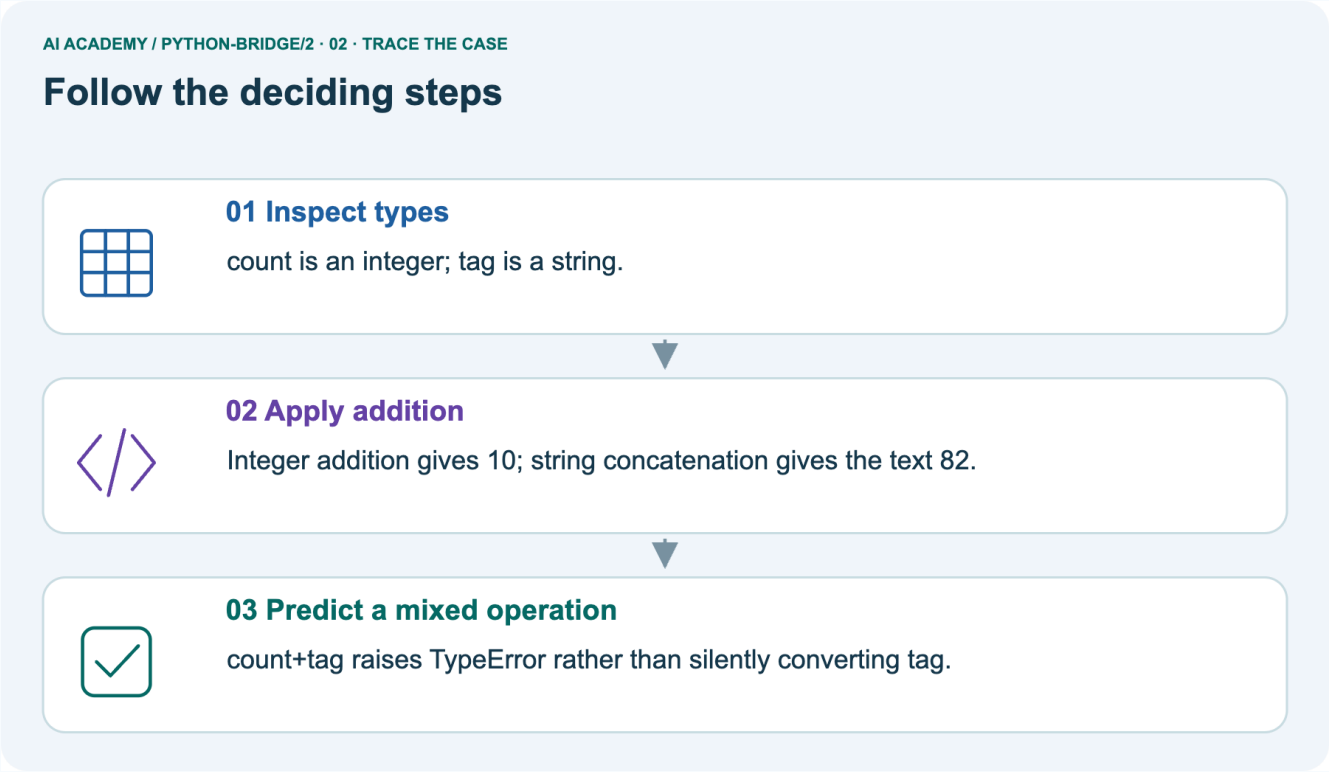
Follow the labelled picture with the matching explanation. Open the vector diagram

A number and its printed-looking text. A program stores a numeric badge count and a **string** that looks like that count. Supplied facts: Python 3: count=8; tag="8"; print(count+2); print(tag+"2").

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Inspect **type**s: count is an integer; tag is a **string**. → Apply addition: Integer addition gives 10; **string** concatenation gives the text 82. → Predict a mixed operation: count+tag raises **Type**Error rather than silently converting tag.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. A saved value is not a live link

AI ACADEMY / PYTHON-BRIDGE/2 · 03 · INSPECT THE LESSON MODEL

A saved value is not a live link

Instruction	count now	saved now
count = 8	8	Not assigned
saved = count	8	8
count = count + 2	10	8

Evaluate the right side using the current value, then bind the left-hand name. The saved integer does not update when count is reassigned.

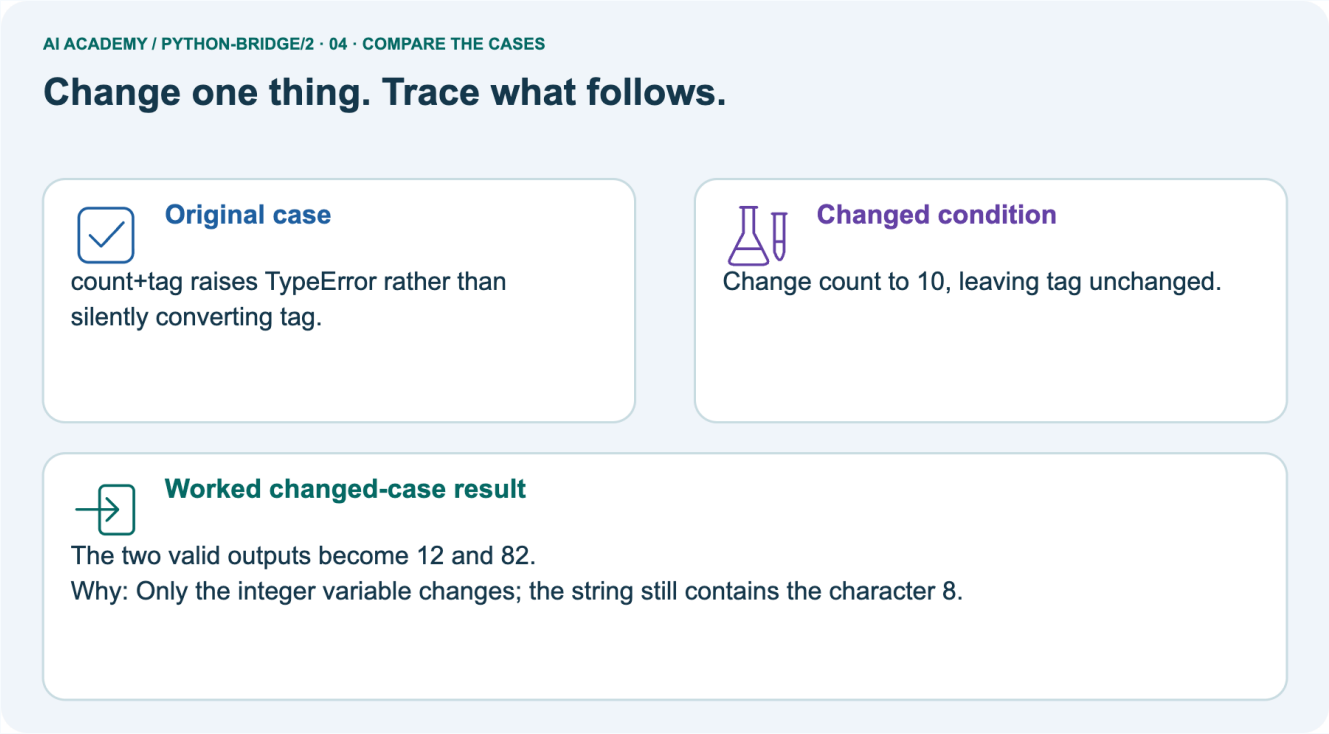
Follow the labelled picture with the matching explanation. Open the vector diagram

A saved **value** is not a live link. count = 8 | 8 | Not assigned. saved = count | 8 | 8. count = count + 2 | 10 | 8

Read the labels and **value**s. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

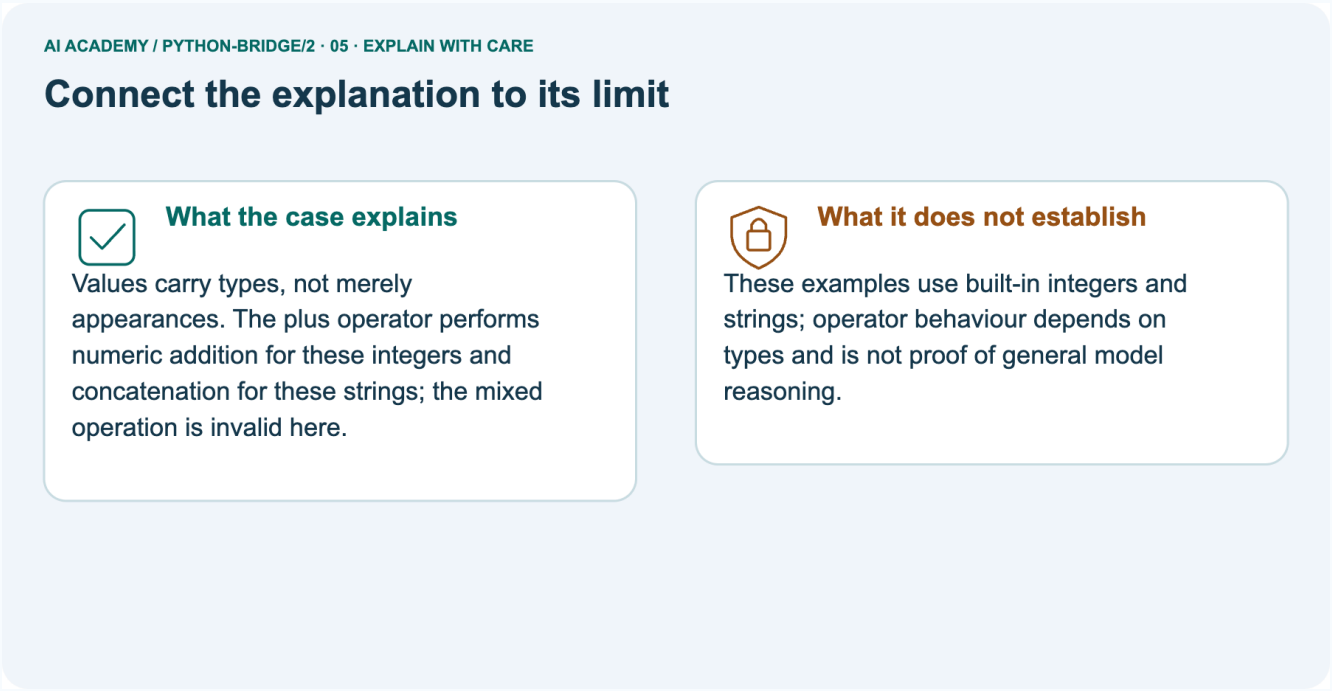
Original case: count+tag raises **Type**Error rather than silently converting tag. Changed condition: Change count to 10, leaving tag unchanged. Expected result: The two valid **output**s become 12 and 82.

Reason: Only the integer **variable** changes; the **string** still contains the character 8.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

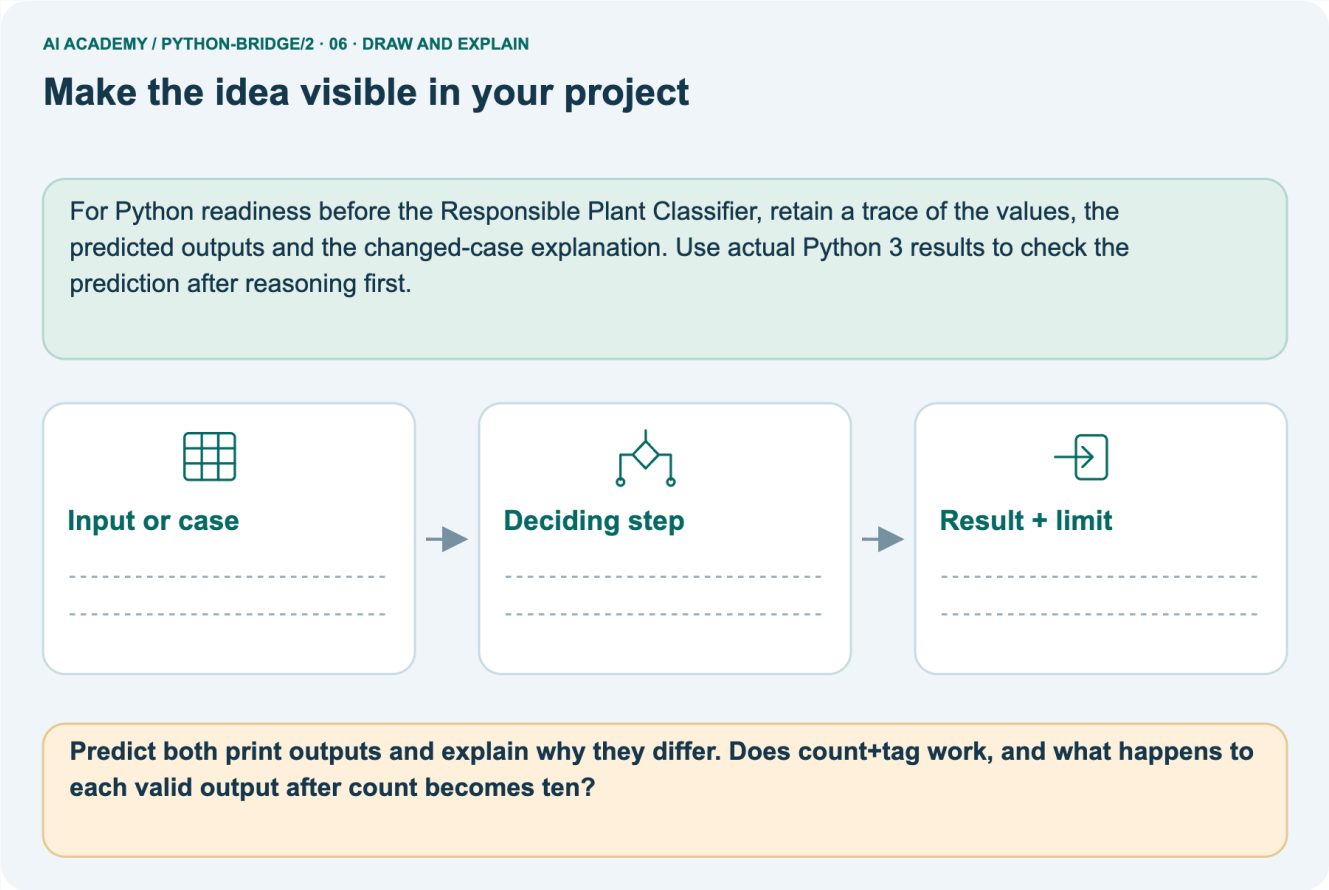
Values carry **type**s, not merely appearances. The plus operator performs numeric addition for these integers and concatenation for these **string**s; the mixed operation is invalid here.

Limit: These examples use built-in integers and **string**s; operator behaviour depends on **type**s and is not proof of general model reasoning.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For Python readiness before the Responsible Plant Classifier, retain a trace of the **value**s, the predicted **output**s and the changed-case explanation. Use actual Python 3 results to check the **prediction** after reasoning first.

Draw your **input**, deciding step, result and limit. Supported question: Predict both print **output**s and explain why they differ. Does count+tag work, and what happens to each valid **output** after count becomes ten?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

01. Values variables and types: the case

02. Follow the deciding steps

03. A saved value is not a live link

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project
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<details>
<summary>Optional interactive practice and completed work</summary>

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.
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